Step-by-Step: Manual Swapfile Configuration

Since resource limits were a major factor in this laboratory environment, I manually provisioned a 2GB swapfile directly within the Parrot OS kernel to serve as a memory safety net. Below is the automated deployment sequence executed inside the terminal:

1. Allocate the swap block (2048MB):
bash
sudo dd if=/dev/zero of=/swapfile bs=1M count=2048

2. Restrict file permissions for system security:
bash
sudo chmod 600 /swapfile

3. Format the file into a Linux swap subsystem:
bash
sudo mkswap /swapfile

4. Engage the swap space into active run-time
bash
sudo swapon /swapfile

5. Append configuration to `/etc/fstab` to ensure persistent mounting across system reboots:
bash
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab

To confirm the operational integrity of the memory buffer, run `free -h` or `swapon --show`.